

Markdown to PDF Converter

A simple and efficient Python tool to convert Markdown files to PDF format using WeasyPrint and the Python Markdown library.

Features

- Convert any Markdown file to PDF with a single command
- Automatic output file naming (same name as input with .pdf extension)
- Custom output file path support
- HTML5 output format for better rendering
- Cross-platform compatibility (Windows, macOS, Linux)

Prerequisites

Before installing the Python dependencies, you need to install system-level packages depending on your operating system:

Ubuntu/Debian

```
bash sudo apt install libpango-1.0-0 libgdk-pixbuf2.0-0 libffi-dev libcairo
```

macOS

```
bash brew install cairo pango gdk-pixbuf libffi
```

Windows

WeasyPrint dependencies are typically handled automatically on Windows, but you may need to install Visual C++ Build Tools if you encounter compilation errors.

Installation

1. Clone this repository: `bash git clone <repository-url> cd md2pdf`
2. Create a virtual environment (recommended): ````bash python -m venv venv`

On Windows:

```
venv\Scripts\activate
```

On macOS/Linux:

```
source venv/bin/activate ````
```

1. Install Python dependencies: `bash pip install -r requirements.txt`

Usage

Command Line Interface

Convert a Markdown file to PDF using the command line:

```
bash python md2pdf.py --filename path/to/your/file.md
```

Optional Parameters

- `--output-file`: Specify a custom output file path `bash python md2pdf.py --filename input.md --output-file custom_output.pdf`

Programmatic Usage

You can also use the `MarkdownPDFConverter` class directly in your Python code:

```
```python from MarkdownPDFConverter import MarkdownPDFConverter
```

## Basic usage

```
converter = MarkdownPDFConverter('example.md') converter.convert()
```

## With custom output file

```
converter = MarkdownPDFConverter('input.md', 'custom_output.pdf')
converter.convert() ```
```

## Examples

### Basic Conversion

```
```bash python md2pdf.py --filename README.md
```

Generates: README.pdf

```
```
```

## Custom Output Location

```
```bash python md2pdf.py --filename docs/guide.md --output-file output/guide.pdf
```

Generates: output/guide.pdf

```
```
```

## Dependencies

- **markdown:** Python library for parsing Markdown syntax
- **weasyprint:** HTML/CSS to PDF converter

## Project Structure

```
md2pdf/ |— MarkdownPDFConverter.py # Core converter class |—
md2pdf.py # Command-line interface |— requirements.txt # Python
dependencies |— Prerequisites.txt # System-level dependencies
|— README.md # This file
```

## Error Handling

The tool includes basic error handling: - Checks if input file exists before processing - Provides clear error messages for missing files - UTF-8 encoding support for international characters

## Supported Markdown Features

This tool supports standard Markdown syntax including: - Headers (H1-H6) - Bold and italic text - Lists (ordered and unordered) - Links - Images - Code blocks - Tables - Blockquotes

## Limitations

- Advanced Markdown extensions may not be fully supported
- PDF styling is basic (no custom CSS support in current version)
- Large files may take some time to process

## Contributing

1. Fork the repository
2. Create a feature branch (git checkout -b feature/amazing-feature)
3. Commit your changes (git commit -m 'Add some amazing feature')
4. Push to the branch (git push origin feature/amazing-feature)

5. Open a Pull Request

## License

This project is open source. Please add your preferred license here.

## Troubleshooting

### Common Issues

1. **WeasyPrint installation errors:** Make sure you have installed the system prerequisites for your operating system.
2. **File not found errors:** Ensure the input file path is correct and the file exists.
3. **Permission errors:** Make sure you have write permissions in the output directory.

### Getting Help

If you encounter issues: 1. Check that all prerequisites are installed 2. Verify your Python environment has the required packages 3. Ensure input files are valid Markdown format

## Version History

- **v1.0.0:** Initial release with basic Markdown to PDF conversion functionality